

Control Approaches for Operating Point Stabilization of Microring Resonator Modulators under Fast Perturbations

Athira Jess Cherian^a, Andreas Michalka^a, Kyle Murray^b, Georg Röhl^b, and Knut Graichen^a

^aChair of Automatic Control, FAU Erlangen-Nürnberg, Germany

^bRanovus GmbH, Nürnberg, Germany

ABSTRACT

This paper presents the design of two different control approaches for stabilizing the operating point of a microring resonator modulator (MRM) when fast disturbances occur. These arise in particular when the properties of the high-frequency modulation signal change due to the rapid electrical and optical self-heating effects. The challenge lies in the relatively slow dynamics of the heater used as an actuator and the significant limitation of the input. The first approach involves the design of a Model Predictive Control (MPC) suitable for a system like MRM. MPC enables a wide operating range by considering the non-linearities of the system. It can operate in both the stable and unstable regions while accounting for constraints on the control input and states of the system. A drawback for real-time implementation, especially in the case of a fast system like the MRM, is that the computational effort is relatively high in each time step due to the included optimization of the control input over a prediction horizon. On one hand, MPC can function as a benchmark design to showcase achievable control quality in simulation, despite its relatively high computational cost. On the other hand, employing a low-order approximation of the dynamic MRM model allows for offline pre-computation of the MPC internal optimization, thereby significantly reducing the online computational effort. The second approach proposes a computationally efficient PID control with two modes. The first mode is designed for normal operation of the PID controller in a region close to the operating point, while the second mode facilitates returning the system back to the operating point (where the PID controller can be restarted) from outside this region, achieved by applying a feedforward control. The system's state variable, ring temperature, determines the operating regions and the corresponding mode selection. To address the challenge of unknown temperature, an Unscented Kalman Filter (UKF) is designed to estimate the temperature.

Keywords: microring resonator modulator, self-heating, MPC, PID, UKF

1. INTRODUCTION

The demand for data traffic and bandwidth is ever-increasing which has made the integrated photonic technologies an area of interest.¹ The optical interconnections realized by silicon photonics are essential for high speed datacenter applications where the need is to overcome the limitations of bandwidth and power-density trade-offs in the high-speed chip-to-chip electrical links.² Microring resonator modulators (MRM) are a crucial part of these interconnections due to their compact footprint and energy efficiency in modulation over other types of modulators.³ Microring resonators are, for instance, used in spectral filters and switches, optical delay lines, bio-sensors, modulators⁴ and as smart sensors via machine learning.⁵ The application of interest in this work is MRM in the transmitter of dense wavelength division multiplexing (DWDM) which encapsulate multiple data streams over closely spaced carrier wavelengths on a single waveguide.⁶ The disadvantage of MRM stems from the fact that perturbations like process variation, change in laser power, laser wavelength, ambient temperature and fast fluctuations, for instance, in the format of the high frequency modulating signal, result in a change in its detuning which may cause more than 3 dB of signal loss for

Further author information: (Send correspondence to A.J.C.)

A.J.C.: E-mail: athira.jess.cherian@fau.de

A.M.: E-mail: andreas.michalka@fau.de

K.M.: E-mail: kyle@Ranovus.com

G.R.: E-mail: georg.roell@Ranovus.com

K.G.: E-mail: knut.graichen@fau.de

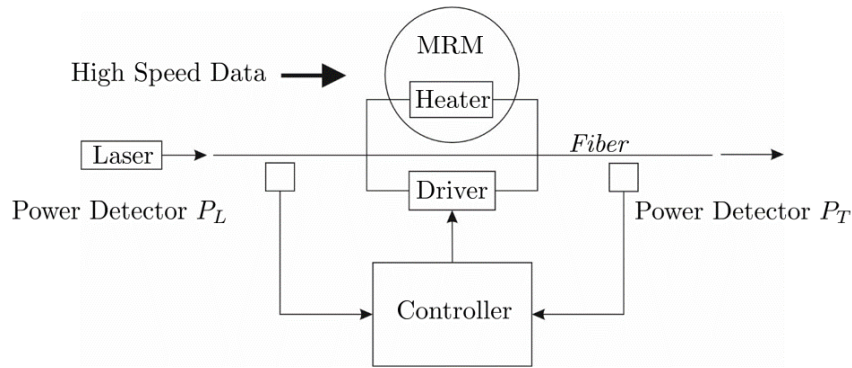


Figure 1: MRM with thermal tuning

a resonance wavelength λ_0 offset of just 0.1 nm (for a silicon microring with a quality factor Q of 10,000 and a nominal resonance wavelength λ_0 in the range of 1180 – 1550 nm).⁷ Silicon microrings experience changes in λ_0 from the manufacturing process variations and temperature fluctuations as in the order of 0.05 – 0.1 nm/K.⁸ When the MRM is driven far-off from the operating point to the undesired region of operation by a sufficiently large thermal fluctuation, the perturbation is aggravated by the optical self-heating effect of MRM (acting as a positive feedback under large input laser power) where the optical power absorbed by the ring also contributes to the heating problem.

The integrated heaters of the MRM can be effectively used to tune them to counteract the process variations due to the strong thermo-optic effect. The same heaters can be utilised as actuators in the control loop strategies to successfully mitigate the perturbations due to the non-process effects which result in the stabilization of operating point. The requirements for designing the control strategies for MRM are: space efficiency and compact footprint, the simplicity in the hardware implementation, less computational effort and fast control to counteract the quick self-heating effects.^{6,9} The challenge involved in the control approach is that the self-heating effect is very fast compared to a slow heater (actuator).

The disturbance considered in this work is the perturbation in temperature due to the fluctuations in the mean voltage of the high frequency modulating signal caused by a sudden change in the patterns or equalization, and the design of a suitable control approach for the operating point stabilization is addressed. Fig. 1 shows the schematic of the MRM with the integrated heater and the controller. The main measurements used for this work are the input power P_L coupled to the microring and the power P_T at the through port which are measured by photodetectors. The insertion loss α , which represents the loss of signal power resulting from the insertion of the microring device in the transmission line, is calculated using these measurements. It is then compared to the target insertion loss α_{OP} and the controller uses the error to adjust the heater power for the thermal tuning.

Since the aim of this work is a suitable control solution to hold the MRM at a constant operating point amidst the thermal disturbances, a model which can capture the influence of thermal effects on the optical power transmitted by the ring is a requirement. In Ref. 20, the microring modulator is analytically modelled in which a time-dependent transfer coefficient is computed by combining the electrical dynamics of the p-n junction of the ring and the optical dynamics. In Ref. 21, the thermo-optic oscillations of the ring resonator have been modelled and the transient response is studied by analyzing the response to a step signal. In Ref. 22, two models relating the physical dimension of the ring to the loss in the ring and the coupling coefficients have been derived. In Ref. 7, an equivalent circuit model of the ring resonator is developed which includes the parasitic effects representing the interconnections, the electrical effects of the p-n junction and the optical effects represented by an LC tank. But the above works do not specifically include the electrical and optical self-heating effects and the relation of the insertion loss to these effects. The model used for this work is based on the linear time invariant system model consisting of the self-heating feedback path described in Ref. 6. In this model, the variation of the resonance wavelength in the presence of perturbation and self-heating is described. In the model developed for this work, the influence of thermal effects arising from the electrical and optical self-heating on the insertion loss



Different control strategies have been discussed in the literature. In Ref. 10, a silicon microring modulator with both an integrated resistive heater and diode-based temperature sensor is demonstrated. Ref. 11 also utilises a temperature sensor for measuring the ring temperature for the feedback loop, but the temperature sensor can increase the space requirement, introduce sensor errors and may itself contribute to the heating problem. Refs. 12 and 13 explain the thermal initialization for wavelength locking which is performed using a dithering signal to generate an error signal for the feedback control. The change in the laser operating wavelength and mean of the modulator voltage can hinder this method from the intended locking. Refs. 14, 15, 16 and 17 utilise the photocurrent in the ring as an error signal for the control loop to adaptively tune the resonance wavelength, but the response time of the loop is slow for a fast perturbation. Refs. 6, 18 present the decoupling of the tracking of optical low and high levels to realize a bit-statistical thermal tuner. The range of wavelength deviation and hence the thermal variations considered in this work is limited. In Ref. 19, the information of the second derivative of the MRM transfer function in a specific working point is extracted using a dithering technique and is used to lock at the point of maximum slope of the transfer function using a controller.

This work is structured as follows. In Section 2, the system models consisting of a heater, optical and electrical self-heating and the nonlinear signal transmission are considered. In Section 3, two control concepts are introduced and evaluated. The discussion begins with a benchmark design using model predictive control to illustrate the achievable control quality. Subsequently, the approach is analyzed for a reduction in computational effort. As a computationally efficient alternative, a PID controller is considered in combination with a switch-back strategy, since the PID itself cannot be used throughout the entire operating range due to the nonlinearities of the system.

Fig. 2 shows the system model consisting of the integrated heater, the optical and electrical self-heating effects and the non-linear signal transmission. The dynamic parts of the system are modeled in simplified form as first-order dynamics. The parameters associated with the model are shown in Table 1.

Table 1: Model Parameters

a. Heater parameters

Param.	V_H	$H_{\Theta HF}$	$H_{\Theta HS}$	τ_{HF}	τ_{HS}	r_0	r_1	r_2
Value	0 - 3.3V	1135K/W	410K/W	13 μ s	78 μ s	122.96 Ω	4.732 Ω/V	4.798 Ω/V^2

b. Microring parameters

Param.	$H_{\Theta SHF}$	$H_{\Theta SHS}$	τ_{SHF}	τ_{SHS}	a	t_1	t_2	R
Value	465K/W	2440K/W	1.7 μ s	7.7 μ s	0.977	0.9135	0.9438	15 μ m

c. Modulator parameters

Param.	$H_{\Theta E}$	τ_E
Value	2905K/W	0.2 μ s

d. Operating point parameters

Param.	$\lambda_{L,OP}$	$P_{L,OP}$	α_{OP}	Θ_{OP}
Value	1308.5nm	10mW	0.25	62.9°C

Signal transmission The transmission of a signal through the microring is primarily influenced by both the modulator signal pattern and the overall temperature of the ring material. These effects cause a phase shift ϕ between the wave that passes through the device directly and the wave circulating within the ring structure, leading to interferences. The phase shift is described by the function:

$$\phi = f_{\phi}(\Theta, \lambda_L) = \frac{4\pi^2 R}{\lambda_L} (n_1 \Theta + n_2 \lambda_L + n_3), \quad (1)$$

with ring radius R , total ring temperature Θ and input laser wavelength λ_L as shown in Fig. 2. The constants n_1 to n_3 are dependent on material properties and dimensions of the MRM and need to be determined through identification. In this work the influence of variations of the laser is not considered, so $\lambda_L = \lambda_{L,OP}$ and $P_L = P_{L,OP}$ (laser power) are assumed to be constant ('OP' denotes the operating point). The phase shift ϕ leads to a drift of the so-called insertion loss α , the ratio of the transmitted power P_T to the input power $P_{L,OP}$:⁴

$$\alpha = \frac{P_T}{P_{L,OP}} = \frac{c_1 - c_3 \cos \phi}{c_2 - c_3 \cos \phi}. \quad (2)$$

As can be seen from eq. (2), a change in ϕ brings about a change in α and therefore in the mean power of the transmitted optical signal. This causes the deterioration of the performance of the transmission system.

Heater The integrated heater within the MRM serves as the actuator, with its input voltage V_H denoting the system's control variable. It is related to the heater power according to:

$$P_H = g(V_H, r_0, r_1, r_2) = \frac{V_H^2}{r_0 + r_1 V_H + r_2 V_H^2}, \quad (3)$$

where r_0 represents the intrinsic resistance of the heater, while r_1 and r_2 describe the static temperature dependency of the resistivity of the heater's material.

The dynamics of the heater are simplifyingly modeled using first-order systems:

$$\dot{\Theta}_{HS} = -\frac{1}{\tau_{HS}}\Theta_{HS} + \frac{H_{\Theta HS}}{\tau_{HS}}P_H, \quad (4)$$

$$\dot{\Theta}_{HF} = -\frac{1}{\tau_{HF}}\Theta_{HF} + \frac{H_{\Theta HF}}{\tau_{HF}}P_H, \quad (5)$$

with the temperature contributions Θ_{HS} and Θ_{HF} , the thermal resistances $H_{\Theta HF}$ and $H_{\Theta HS}$ and the time constants τ_{HF} and τ_{HS} of the fast and slow components of the heater, respectively.

Optical self-heating The first component of self-heating is caused by optical effects. The optical power P_A absorbed in the microring resonator is related to the input power $P_{L,OP}$ by:⁴

$$\gamma = \frac{P_A}{P_{L,OP}} = \frac{c_2 - c_1 - c_4}{c_2 - c_3 \cos \phi}, \quad (6)$$

with $c_1 = t_1^2 + t_2^2 a^2$, $c_2 = 1 + t_1^2 t_2^2 a^2$, $c_3 = 2t_1 t_2 a$ and $c_4 = (1 - t_1^2)(1 - t_2^2)a$, where t_1, t_2 are the coupling coefficients and a is the round trip loss coefficient of the ring.

The dynamics of the optical self-heating are also simplifyingly modeled using first-order systems:

$$\dot{\Theta}_{SHS} = -\frac{1}{\tau_{SHS}}\Theta_{SHS} + \frac{H_{\Theta SHS}P_{L,OP}}{\tau_{SHS}}\gamma, \quad (7)$$

$$\dot{\Theta}_{SHF} = -\frac{1}{\tau_{SHF}}\Theta_{SHF} + \frac{H_{\Theta SHF}P_{L,OP}}{\tau_{SHF}}\gamma, \quad (8)$$

with the temperature contributions Θ_{SHS} and Θ_{SHF} , the thermal resistances $H_{\Theta SHF}$ and $H_{\Theta SHS}$ and the time constants τ_{SHF} and τ_{SHS} of the fast and slow components of optical self-heating, respectively.

Electrical self-heating The second component of self-heating is caused by the electrical effect of the modulator signal and is considered as a disturbance for the control system to be designed. In this case as well, the dynamics are modeled by a first-order system:

$$\dot{\Theta}_E = -\frac{1}{\tau_E}\Theta_E + \frac{H_{\Theta E}}{\tau_E}P_{MOD}, \quad (9)$$

with the temperature contribution Θ_E , the thermal resistance $H_{\Theta E}$ and the time constant τ_E of the electrical self-heating model. A change in the transmitted signal pattern is reflected as a change in the mean power P_{MOD} of the modulator leading to thermal fluctuations which are further escalated by the optical self-heating effects of the ring on the unstable side of the resonance.

System output and total temperature The insertion loss α as described in eq. (2) is considered as the system output and thus as the controlled variable. As shown in Fig. 2 the variables Θ_{HS} , Θ_{HF} , Θ_{SHS} , Θ_{SHF} , and Θ_E represent the temperature contributions of the slow and fast parts of the heater, the slow and fast parts of optical self-heating, and the electrical self-heating, respectively. Together with the ambient temperature Θ_0 , they sum up to the total temperature

$$\Theta = \Theta_{HS} + \Theta_{HF} + \Theta_{SHS} + \Theta_{SHF} + \Theta_E + \Theta_0, \quad (10)$$

which is an important variable of the system and describes the current operating point.

Simplified system dynamics To reduce computational effort, a simplified model of the system dynamics can be useful for controller design, assuming a single state variable Θ_Σ , for which the dynamics follow as:

$$\dot{\Theta}_\Sigma = -\frac{1}{\tau_\Sigma}\Theta_\Sigma + \frac{H_{\Theta H}}{\tau_\Sigma}P_H + \frac{H_{\Theta SH}P_{L,OP}}{\tau_\Sigma}\gamma + \frac{H_{\Theta E}}{\tau_E}P_{mod} =: f(\Theta_\Sigma, V_H, P_{MOD}), \quad (11)$$

where τ_Σ is the approximated time constant of the system obtained using optimization for the purpose of simplification as derived from eqs. (4), (5), (7) and (8). Taking eqs. (3), (6) and (1) into account, the temperature dynamics can be described as a nonlinear function f . The thermal resistances are likewise combined accordingly, i.e. $H_{\Theta H} = H_{\Theta HS} + H_{\Theta HF}$ and $H_{\Theta SH} = H_{\Theta SHS} + H_{\Theta SHF}$.

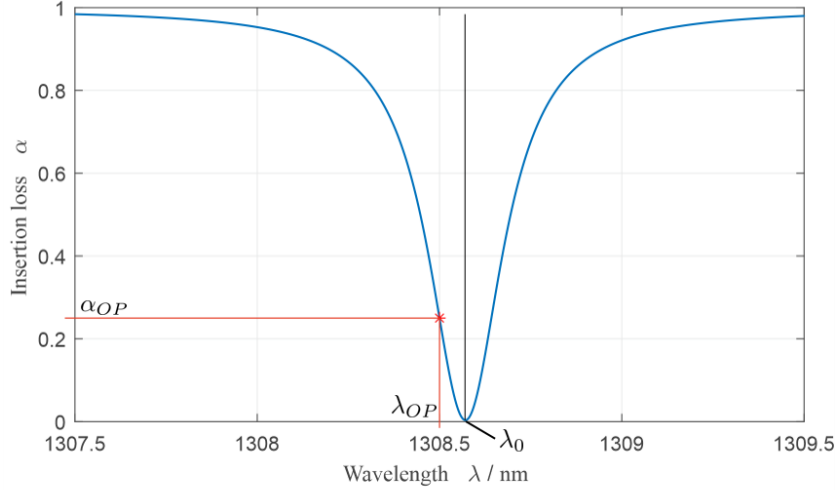


Figure 3: Transfer characteristics of the MRM

3. CONTROL DESIGN AND RESULTS

In this section, two different controller designs are outlined. As a first control approach, Model Predictive Control (MPC) is employed, where the nonlinearities of the system are taken into consideration. This control approach suggests that operating points can be stabilized throughout the entire operating range of the system, despite the nonlinearities. Implementing MPC for very fast systems like the MRM is a significant challenge; the optimization involved in the process demands a highly efficient implementation. Nevertheless, MPC can serve as a benchmark design to evaluate the performance of other designs. For simple system models, there is also the option to perform the optimization offline in advance and store the results, as demonstrated in this work. The second approach is a classical PID design, for which an implementation with low computational effort can be expected. On the other hand, the range in which the operating point can be stabilized will be limited due to the nonlinearities of the system that are not considered at that point. Additional components may be necessary in the control system to address this situation.

Control task The transfer characteristics (2) of the ring for a single mode of operation⁶ is shown in Fig. 3. The resonance wavelength λ_0 corresponds to the point $\alpha = 0$, where the power gets fully absorbed in the microring (in the ideal model) due to the constructive interference. This is the boundary of the two regions of operation marked by the dashed line. In the so-called blue-biased region ($\lambda_0 < \lambda_{L,OP}$, right side of the dashed line in Fig. 3) operating points are unstable under sufficiently large input power P_L , and in the so-called red-biased region ($\lambda_0 > \lambda_{L,OP}$, left side of the dashed line in Fig. 3) operating points are stable thus leads to a desired operation of the MRM in this region. Physically interpreted, a red-biased ring ($d\alpha/d\lambda < 0$) implies negative feedback in the self-heating effect. In the blue-biased case ($d\alpha/d\lambda > 0$), the self-heating feedback is positive. Any perturbation to the ring is continuously amplified by self-heating, leading to a drift away from the operating point in ring resonance.

The task of the controller is to maintain the average optical power transmitted at a fixed operating point, i.e., retaining the system at a constant α . In Fig. 3, a possible operating point ($\lambda_{L,OP}, \alpha_{OP}$) is marked with an asterisk. The definition of the operating point includes the corresponding effective heating power introduced into the microring by a nominal signal pattern. Any variation in this heating power P_{MOD} when transmitting different signal patterns can introduce disturbances to the lower-frequency temperature behavior, thereby affecting the operating point of the control under consideration. A further important objective for the controller design is to keep the system, if possible, in the stable red-biased region regardless of fluctuations.

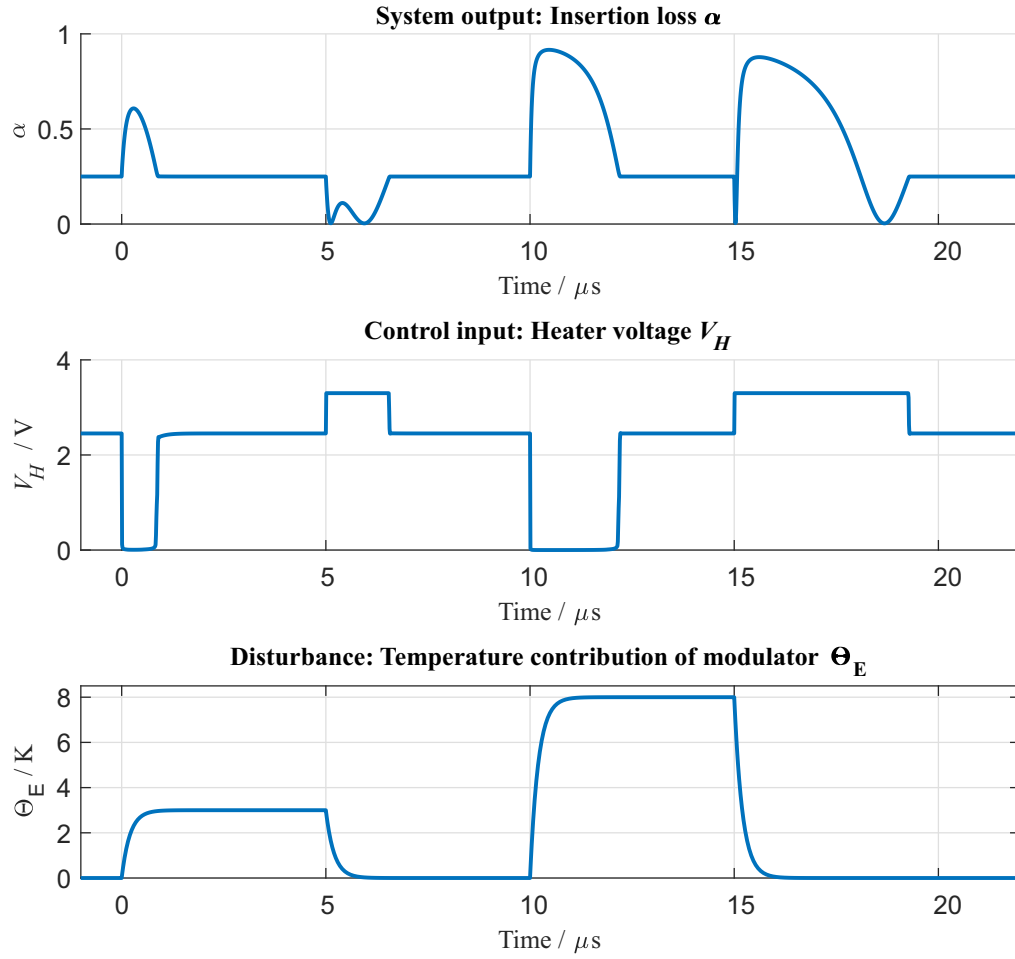


Figure 4: MPC results with 4 different disturbance scenarios

3.1 Model Predictive Control

A modern control method suitable for nonlinear systems is model predictive control (MPC).²⁴ With the knowledge of the state of the system at the current time point, MPC determines an optimized control input signal for a prediction horizon through the solution of a cost minimization problem. This optimal solution is then applied as the control input over the current sampling interval, and the minimization problem is solved again in the next sampling step. The MPC formulation of this problem is based on the system model described in Section 2 using the simplified system dynamics (11). The temperature corresponding to the desired setpoint is $\Theta_{\Sigma,OP}$, and for the initial state holds $\Theta_{\Sigma,0} = \Theta_{\Sigma,OP}$. MPC serves as the controller for the system based on the solution of the optimal control problem as follows:

$$\min_{\bar{V}_H(\cdot)} J(\Theta_{\Sigma,k}, \bar{V}_H) = \int_0^{T_h} l(\bar{\Theta}_{\Sigma}(\tau), \bar{V}_H(\tau)) d\tau \quad (12a)$$

$$\text{s.t.} \quad \dot{\bar{\Theta}}_{\Sigma}(\tau) = f(\bar{\Theta}_{\Sigma}(\tau), \bar{V}_H(\tau)), \quad \bar{\Theta}_{\Sigma}(0) = \Theta_{\Sigma,k} = \Theta_{\Sigma}(t_k) \quad (12b)$$

$$\bar{V}_H(\tau) \in [0, 3.3V], \quad \tau \in [0, T_h], \quad (12c)$$

with the current system state $\Theta_{\Sigma,k}$ at the time point t_k ($t_k = kT_s$, where k is the sampling instant and T_s is the sampling time of the control system). The prediction horizon T_h defines the time interval $[0, T_h]$ on which the

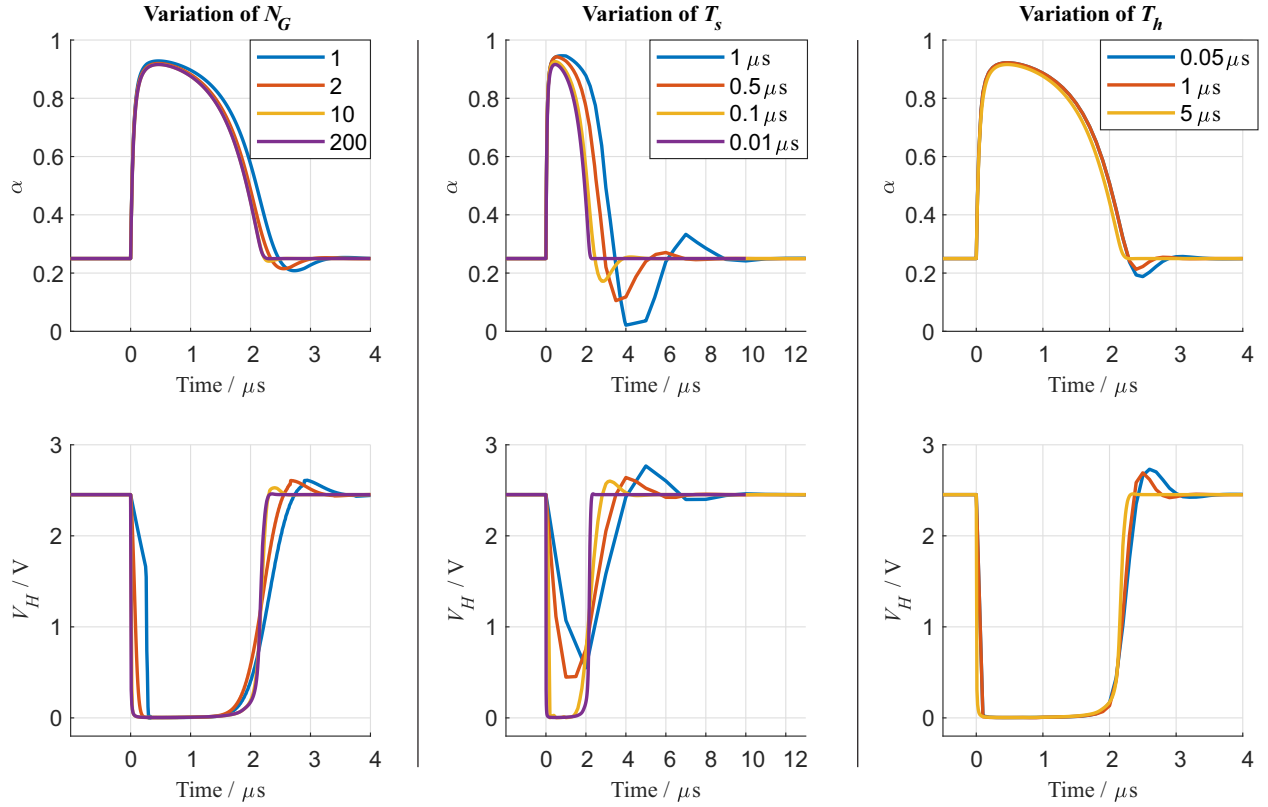


Figure 5: Effect of parameter variation

system trajectories are predicted. The bar indicates MPC-internal variables with the prediction time coordinate $\tau \in [0, T_h]$ in contrast to the real system dynamics (11) with time $t \geq 0$. The integral cost function in (12a) is chosen as:

$$l(\Theta_\Sigma, V_H) = Q \cdot (\Theta_\Sigma - \Theta_{\Sigma,OP})^2 + R \cdot (V_H - V_{H,OP})^2, \quad (13)$$

where the deviations of the state Θ_Σ and the control input V_H from the operating point are penalized by weighting the errors with positive constants Q and R , respectively.

The solution of (12) and thus the resulting MPC-internal optimal control input $\bar{V}_H^*(\tau)$ for the prediction horizon at time instant t_k determines the control input of the system for the current sampling interval ($t \in [t_k, t_{k+1})$):

$$V_H(t_k + \tau) = \bar{V}_H^*(\tau), \quad \tau \in [0, T_s]. \quad (14)$$

In the subsequent sampling step $t_{k+1} = t_k + T_s$, the problem (12) is re-solved with the new initial state $\Theta_{\Sigma,k+1}$, giving rise to a moving horizon concept. In practice, a discretization is commonly employed to keep $V_H(t_k + \tau)$ constant over the sampling interval.

The MPC approach used in this work is implemented through the software framework GRAMPC.²⁵ The important parameters affecting the control quality are the sampling time T_s , the prediction horizon T_h , the number of gradient iterations N_G (the number of the gradient iterations in the gradient method used for solving the inner minimization problem) and the weights of the states Q and the input R in the cost function. For application it is required to strike a balance between the computational effort and the speed of compensation of a fast disturbance, depending on the performance of the available hardware.

However, as a first step, MPC is intended to be used as a benchmark design in simulation. This means that the parameters are chosen without consideration for computational effort and real-time capability, but for a fast control behaviour, aiming to achieve the best possible result: $N_G = 200$, $T_s = 0.01 \mu s$, $T_h = 5 \mu s$, $Q = 1000$ and

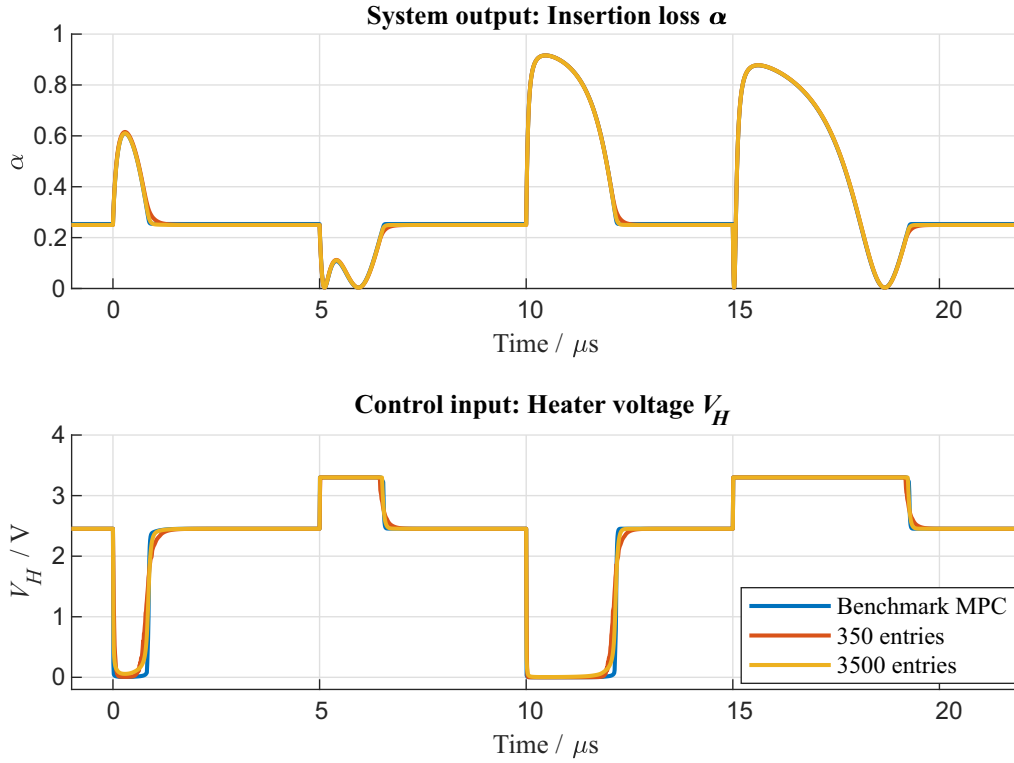


Figure 6: Effect of pre-calculation of MPC control inputs

$R = 1$. In a second step, the simplification of MPC to enhance computational efficiency will be considered.

Fig. 4 shows the results of a benchmark MPC design, considering four different disturbance scenarios. At $0\mu\text{s}$, $5\mu\text{s}$, $10\mu\text{s}$, and $15\mu\text{s}$, step shaped disturbances from the modulator voltage bringing a change in the overall temperature Θ of 3K, -3K , 8K and -8K are applied. In all 4 scenarios, the MPC controller can bring the system back to the operating point. Initially, caused by the large disturbances and due to the limitation of the control input, significant deviations occur, accentuating the nonlinearities of the system. Even in Scenario 4, where a transition to the unstable (red-biased) region occurs, the controller can stabilize the operating point.

This consideration clearly demonstrates the broad operating range of the approach, resulting from the incorporation of the nonlinear system model in the design. Furthermore, stabilization of the operating point is achievable despite the heavily constrained control signal, as these constraints are also taken into account in the control approach. If the question of implementation arises, however, a high computational effort, especially in systems with rapid dynamics like the MRM, poses a challenge.

To reduce the computational effort, the parameterization of the MPC can be adjusted. However, this has negative effects on the control quality, as shown in Fig. 5. It is evident that the result of the minimization problem contained in the MPC becomes suboptimal. However, the computation time of 2.9ms for one sampling step of the benchmark MPC on a standard simulation PC (Core i7) can be reduced. For example, by setting N_G to 1, the time can be decreased to $20\mu\text{s}$, with T_s set to $1\mu\text{s}$, the time can be reduced to $250\mu\text{s}$, and with T_h set to $0.05\mu\text{s}$, the time can be decreased to $250\mu\text{s}$, all while keeping the other parameters constant, respectively.

Significantly greater potential lies in the offline pre-calculation of the minimization problem contained in the MPC. Here, it is possible to calculate with accuracies at the level of the benchmark simulation and save the results for online use. However, this is only possible for low-order systems, as otherwise the combinations of state values to be captured lead to an explosion of memory requirements. This approach is shown below for the simple model used here.

Pre-calculation of MPC The online computation of the internal minimization problem of MPC leads to complex calculations due to the computation of gradients, gradient iterations and repeating the process for as many prediction steps as described by the prediction window length and the sampling time. To compensate for fast perturbations for the considered system, the control input needs to be computed within sub-microseconds. Therefore it is impossible to calculate the control input online within the short sampling interval using the MPC algorithm as there is limitation in the currently available computer technology.

With the application of the first-order approximation of the system model (refer to eq. (11)), the MPC design has been simplified into a one-state variable problem. This allows for the pre-calculation and storage of optimization results across the entire state space in a lookup table without generating an immense storage requirement as in the case of multiple states. During control operation, the required control input V_H could be retrieved from the lookup table depending on the present system temperature Θ . Another advantage is that the benchmark parameter values for N_G , T_s , and T_h can be used for pre-computation of the control input without compromising the reduction in computational burden. Three cases of comparison are shown in Fig. 6 using the same disturbance scenarios as shown in Fig. 4. Case 1 is the benchmark MPC design that is computed online, case 2 refers to a pre-calculation MPC with a look-up table resolution of 0.1 °C (350 entries), and case 3 is a pre-calculation MPC with a look-up table resolution of 0.01 °C for Θ_Σ in the range of 55 - 90 °C (3500 entries). If the lookup table with 350 entries is used, slight compromises in control quality can be expected. With the more precise lookup table with 3500 entries, the results are nearly identical compared to the benchmark MPC.

The considered MRM represents a system with very fast dynamics and, simultaneously, a strongly limited control input. The results of the MPC show that the control signal frequently matches the limitation value for an extended period when responding to fast disturbances. Therefore, in the following, we will investigate whether a controller design with less implementation effort can achieve similar results. For this purpose, we will consider a classical PID design.

3.2 PID control

The classical PID approach has the advantage that it is easy to implement in hardware and is flexible and efficient in terms of footprint. The structure of the controller with the system is shown in Fig. 7. The output of the controller is the requested change in heater voltage $\Delta V_{H,C}$. The design of PID control for the microring resonator begins with the tuning of controller gain parameters k_p, k_i, k_d . In this work, it is achieved through optimization by simulating the model under the worst-case disturbance, realized as a change in modulator power ΔP_{MOD} , inducing a step change in the system temperature with an amplitude of approximately 8 K. The optimized gains will be the ones resulting from the minimised value of the following objective problem:

$$\begin{aligned} \min_{\Delta V_H} \quad & w_1 \cdot \Delta \alpha^2 + w_2 \cdot \Delta V_H^2 \\ \text{s.t.} \quad & \text{Model equations (1) – (10),} \\ & 0 \leq V_H \leq 3.3V. \end{aligned} \tag{15}$$

In this optimization, special attention is given to ensuring that, in the presence of disturbances, the deviation $\Delta \alpha$ remains as small as possible to avoid transitioning into the unstable region. As a first step, as in eq. (15), the weights w_1, w_2 of the deviations $\Delta \alpha, \Delta V_H$ should be selected suitably. In this context, only the ratios of the weights to each other, but not the values themselves, are crucial for the optimization result. In Fig. 8, a worst case scenario with a step-shaped disturbance affecting a temperature deviation of 8 K is considered. Four different example cases of weight ratio choices are shown, $w_1/w_2 \in \{0.005, 0.5, 500, 5000\}$. It is obvious that if the weight ratio w_1/w_2 is increased, the optimization results in a higher controller gain and therefore a higher ΔV_H with a better utilization of the control input signal range.

If this controller is to be implemented on a digital computer, the required sampling time is also a crucial factor, as it is with MPC, as a controller with higher gain and faster controlled system's dynamics requires a smaller sampling time. Thus it is always a trade-off between the settling time and the available computational hardware. The cases shown in Fig. 8 are simulated using a digital controller model, with the sampling time individually maximized in each case. The resulting required sampling times, $T_s = \{10\mu s, 5\mu s, 1\mu s, 0.5\mu s\}$, highlight the significant difference in computational effort for the implementation of the PID controller, depending on the chosen parameter values.

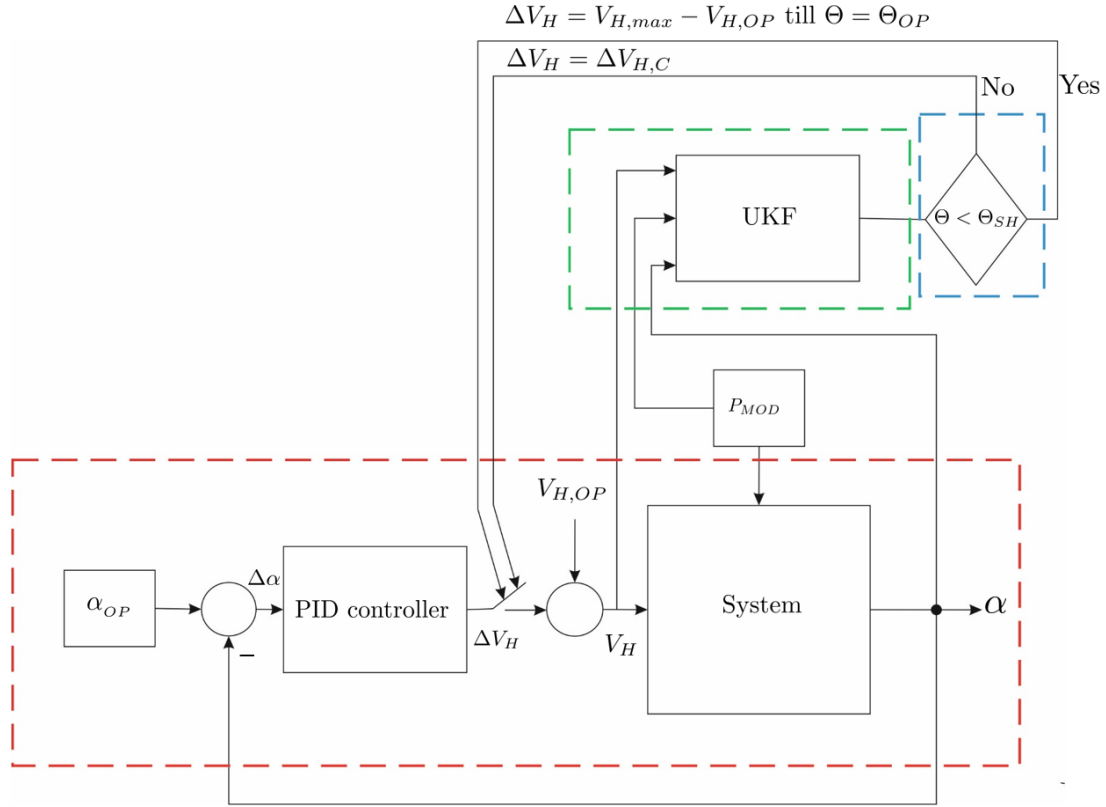


Figure 7: PID Control Approach with UKF Temperature Estimation and Switch Back Mode

In Fig. 4 a comparison of the MPC results (blue curves) discussed in section 3.1 and the PID results with $w_1/w_2 = 5000$ (red curves) is presented. In the first three cases, the PID controller can compensate for the disturbance and bring the system back to the operating point. The signals are similar to those of MPC, demonstrating that equivalent control quality can be achieved for this special case of a fast dynamical system with strong limitations on the control input using both the MPC and PID approaches. In case 4, however, the PID controller fails to bring the system back to the operating point, and the system enters the unstable region. As the PID controller, by its conceptual nature, cannot return the system from this region, the controller needs to be deactivated. This necessitates an additional mode, referred to as the 'switch-back mode', which can restore the system from the unstable region.

Switch-back mode with UKF

To determine whether the operating range of the PID controller has been exceeded and whether it needs to be deactivated, as well as for the appropriate return to the operating point, current information about the system state is necessary. This information is provided by the overall temperature of the microring, which, however, is not measurable. Hence, the temperature is estimated, and due to the strong nonlinearities of the system, especially in the transition area between the red- and blue-biased regions (see Fig. 3), an Unscented Kalman Filter (UKF),²³ a method developed specifically for nonlinear systems, is employed (see Fig. 7). The UKF concept is based on a discrete-time model (sampling time T_s) of the system in the form:

$$\mathbf{x}_{k+1} = \mathbf{F}(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{w}_k, \quad (16)$$

$$y_k = H(\mathbf{x}_k, \mathbf{u}_k) + v_k, \quad (17)$$

where $\mathbf{x}_k$ represents the states (not measured) of the system at the k^{th} sampling instant. The vector $\mathbf{u}_k$ contains the known input signals, and the output variable y_k is the only measured signal. For the UKF design, the effects

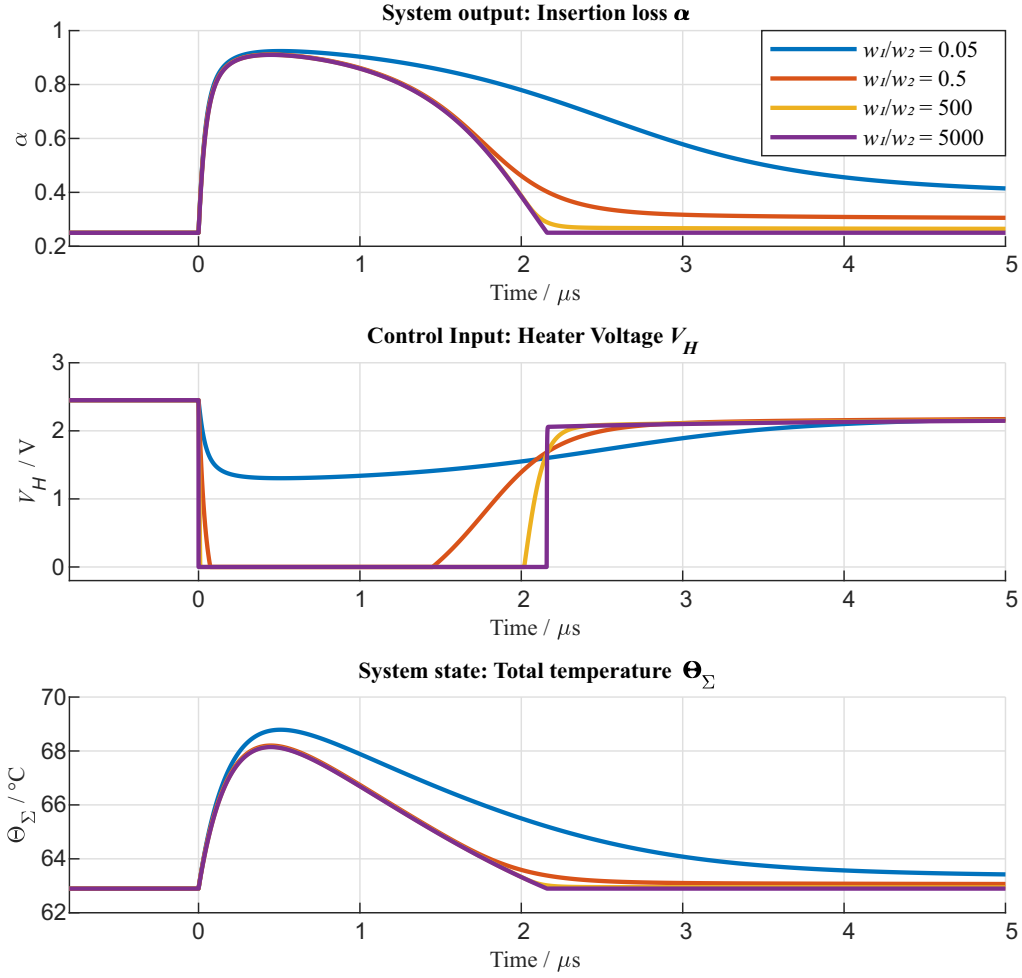


Figure 8: Optimization-based PID controller design with different weight choices

of disturbances in the system are described through the process noise $\mathbf{w}_k$ and the measurement noise v_k , both are assumed as Gaussian random variables (GRV).

In a first step the UKF evaluates the model to predict the output value $\hat{y}_{k+1}^-$ at the next sampling instant, which corresponds to a propagation of a GRV through the non-linear system dynamics. To implement this with high accuracy, a set of carefully chosen sample points known as sigma points are used which are found performing the unscented transformation around the mean. They capture the true mean and covariance of the GRV.²³ In the second step of the estimation, the predicted state variable is corrected by comparing the measured output signal with the predicted output, yielding the estimated values for the states $\hat{\mathbf{x}}_{k+1}$.

The UKF design for the MRM relies on the simplified system dynamics model (11) with only one state variable. As the UKF concept is based on a discrete-time model of the considered system, eq. (11) can be transformed into the discrete-time domain using the Euler approximation, resulting in the state transition function F according to eq. (16). During the prediction phase, every sigma-point undergoes the non-linear transformation using the state transition function as follows:

$$\Theta_{\Sigma,k+1} = \Theta_{\Sigma,k} + T_s f(\Theta_{\Sigma,k}, V_{H,k}, P_{MOD,k}) =: F(x_k, \mathbf{u}_k), \quad (18)$$

where the UKF estimates the state of total temperature $\hat{\Theta}_\Sigma$ of the system.

The discrete-time output equation follows by combining eqs. (2) and (1) and evaluating at the k^{th} sampling

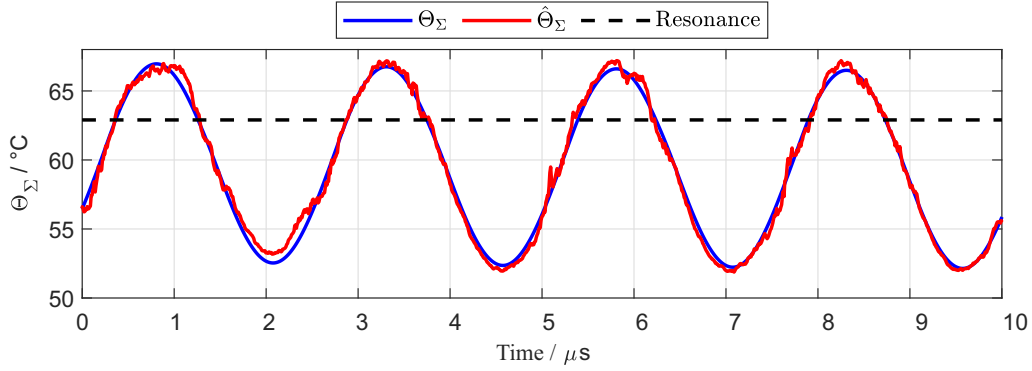


Figure 9: Temperature estimation using UKF

instant. The estimated output is calculated for each of the sigma-points using the measurement function H described in eq. (17) as follows:

$$\alpha_k = \frac{c_1 - c_3 \cos\left(\frac{4\pi^2 R}{\lambda_{L,OP}} (n_1 \Theta_{\Sigma,k} + n_2 \lambda_{L,OP} + n_3)\right)}{c_2 - c_3 \cos\left(\frac{4\pi^2 R}{\lambda_{L,OP}} (n_1 \Theta_{\Sigma,k} + n_2 \lambda_{L,OP} + n_3)\right)} =: H(x_k, \mathbf{u}_k). \quad (19)$$

During the correction phase, the estimated output is compared to the measured output of the real system supplied to the UKF and the corrected estimate of the mean and covariance of the state is returned. The main parameters involved in the design of UKF are the spread of the sigma points around the mean, the scaling parameter, the initial covariance of the state and the sampling time. The choice of parameters like sampling time are a compromise between the estimation error and computational effort.

Fig. 9 illustrates the temperature estimation with the UKF when a sinusoidal thermal fluctuation, including measurement noise, is applied. This fluctuation drives the system through both the stable and unstable regions, which are separated by the resonance point. It demonstrates the estimation quality of the UKF.

The estimated temperature $\hat{\Theta}_{\Sigma}$ facilitates the decision to deactivate the PID controller upon reaching the threshold Θ_{SH} (see Fig. 7), triggering the start of the switch-back mode. In this mode, the upper limit of the control input V_H is applied until the estimated temperature aligns with the desired operating point. Subsequently, the PID controller is reinitialized.

Fig. 10 shows the response of the system with the controller and the switch-back mode to the same disturbance scenarios as earlier. At $15 \mu s$, when the system with only feedback mode could not bring back the system to the stable region and to the setpoint, the proposed system with the switch-back mode along with UKF could handle the disturbance well and brings the system back to the setpoint. Therefore it can be seen that the thermal fluctuations can be compensated well in both directions by an enhancement of classic PID controller in this way. Compared to the MPC, apparently similar behavior is observed (see Fig. 5), which is reasonable since the control input is almost always in the constraints.

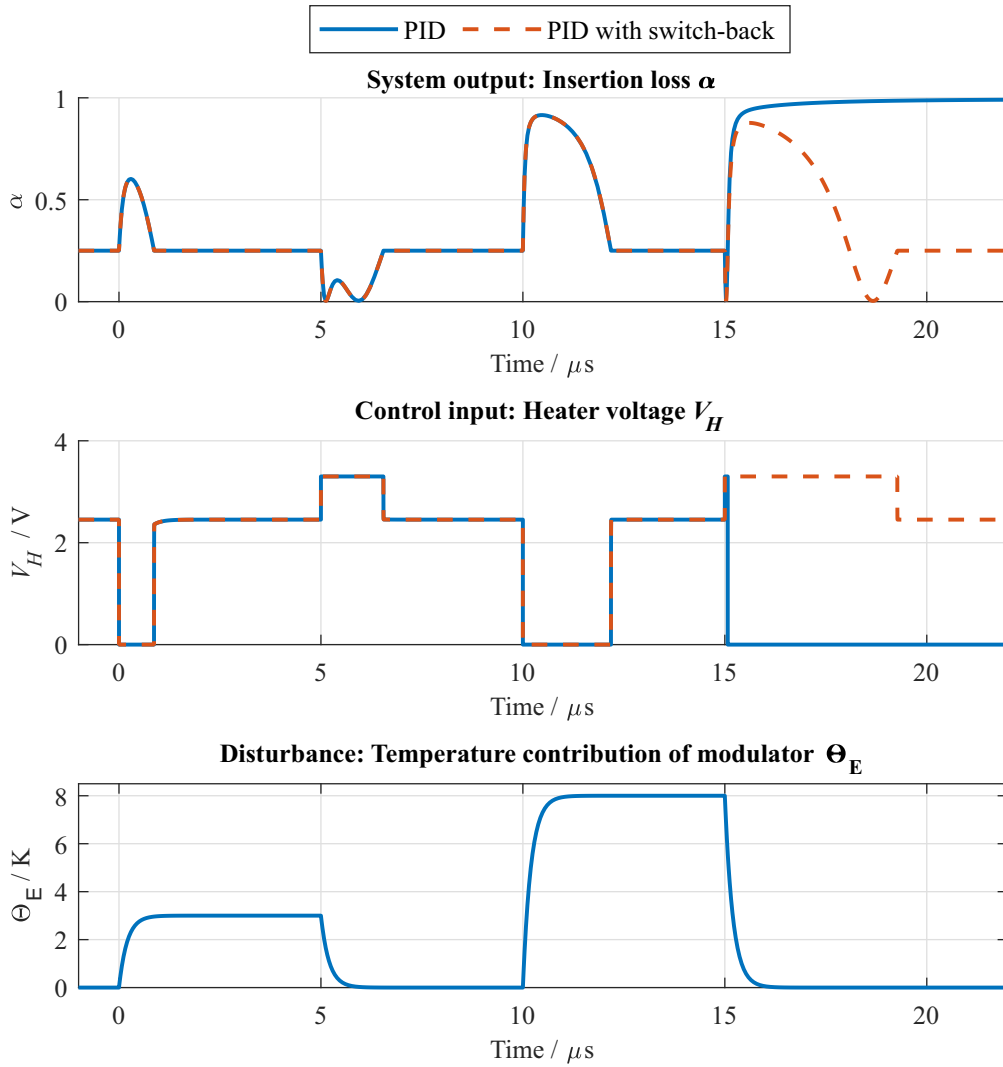


Figure 10: Comparison of PID and PID with switch-back mode

4. CONCLUSION

In this work, two different control approaches to mitigate the effect of thermal perturbations in a microring resonator modulator (MRM) due to very rapid changes in the modulation pattern have been discussed. In the first approach, a modern control method, Model Predictive Control (MPC), has been designed. Its primary purpose is to serve, on the one hand, as a benchmark control to showcase the potential of the controlled system. Due to the non-linear system model MPC can provide a broad operating range. On the other hand, it was demonstrated that MPC design can be adapted to reduce computational effort for a potential real-time hardware implementation. A significant advantage lies in the pre-calculation of the internal minimization included in the MPC approach, offering a promising option for low-order systems. In the second approach, a classical PID control with an additional switch-back mode utilizing temperature estimation with UKF has been proposed. Comparable control quality to the MPC approach was achieved for the MRM, a system characterized by rapid dynamics and strongly limited control input. As this concept lacks optimization, advantages in terms of computation time arise from this.

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